



Institute of Information Technology
University of Dhaka

SE801 • SOFTWARE PROJECT LAB 3 • FINAL DEFENSE

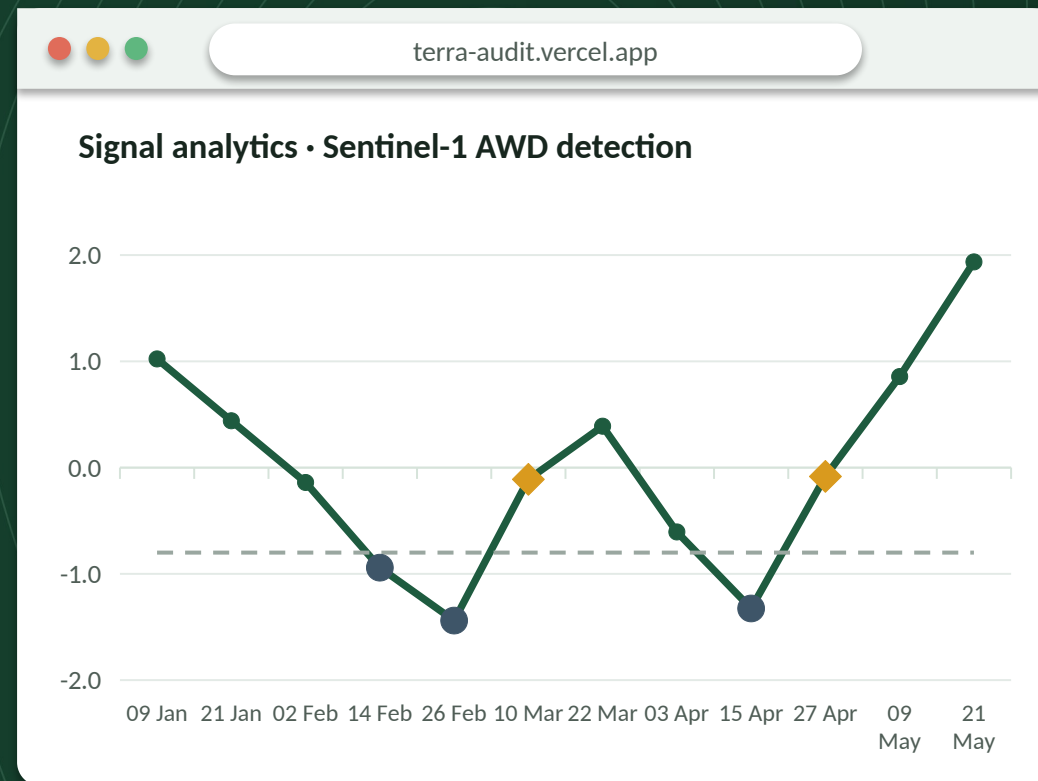
Terra-Audit

AI-Assisted Digital MRV Platform for AWD Rice Irrigation
and Agricultural Land Management Carbon Credits

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DATE
October 2026
Final defense



2 AWD drydowns
→ 2.12 tCO₂e / ha

Verra VM0051 • VM0042

Flooded rice emits methane — and AWD reductions are hard to prove

METHANE EMISSION FACTOR

1.00 → 0.55

Alternate Wetting & Drying (AWD) —
draining the field at least twice per season
— cuts the daily methane factor by 45%
versus continuous flooding.

IPCC 2019 Table 5.12, used by Verra VM0051



Practice claims are hard to verify

Farmer irrigation logs are incomplete; sensors on every smallholder plot are too costly.



Evidence gets separated from the number

Spreadsheets lose the factors, methodology version and history behind a result.

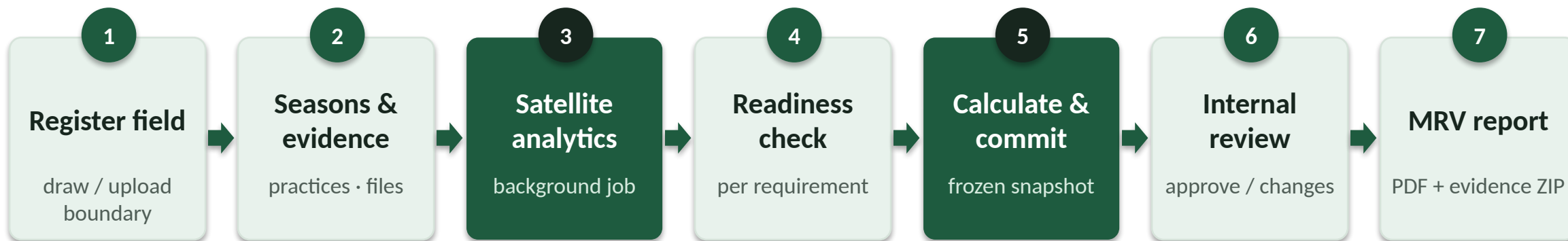


Verification needs traceability

Verifiers and buyers must trace each tonne of CO₂e back to its evidence.

Terra-Audit links every carbon number to its evidence

An evidence-first digital MRV platform — from field boundary to a verifier-ready MRV report.



Rice — Alternate Wetting & Drying

Verra VM0051 v1.1

Satellite-detected drydowns cut methane; 15% QA3 uncertainty and the N₂O penalty are deducted.

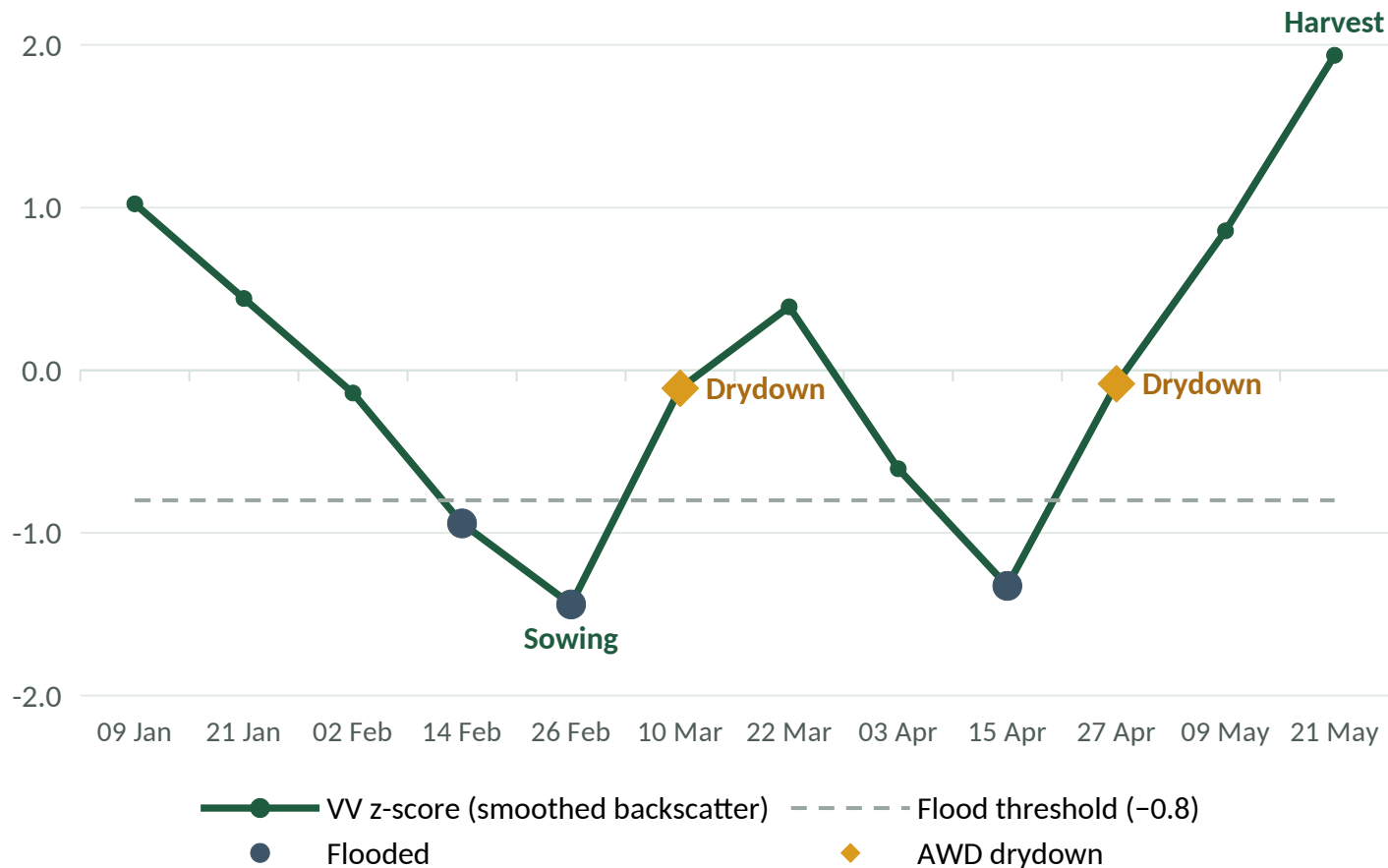


Agricultural Land Management

Verra VM0042 v2.2 + VMD0054 v1.1

Measured soil-carbon change plus N₂O, fire and fuel terms, minus leakage and a non-permanence buffer; missing leakage evidence blocks the result.

Sentinel-1 radar detects AWD drydowns without field sensors



Flooded

VV z-score below -0.8

Drydown

VV rises $> 1.2\sigma$ right after a flooded date

Season

VH minimum = sowing; sharpest VH drop after the peak = harvest

Result fed to the calculation

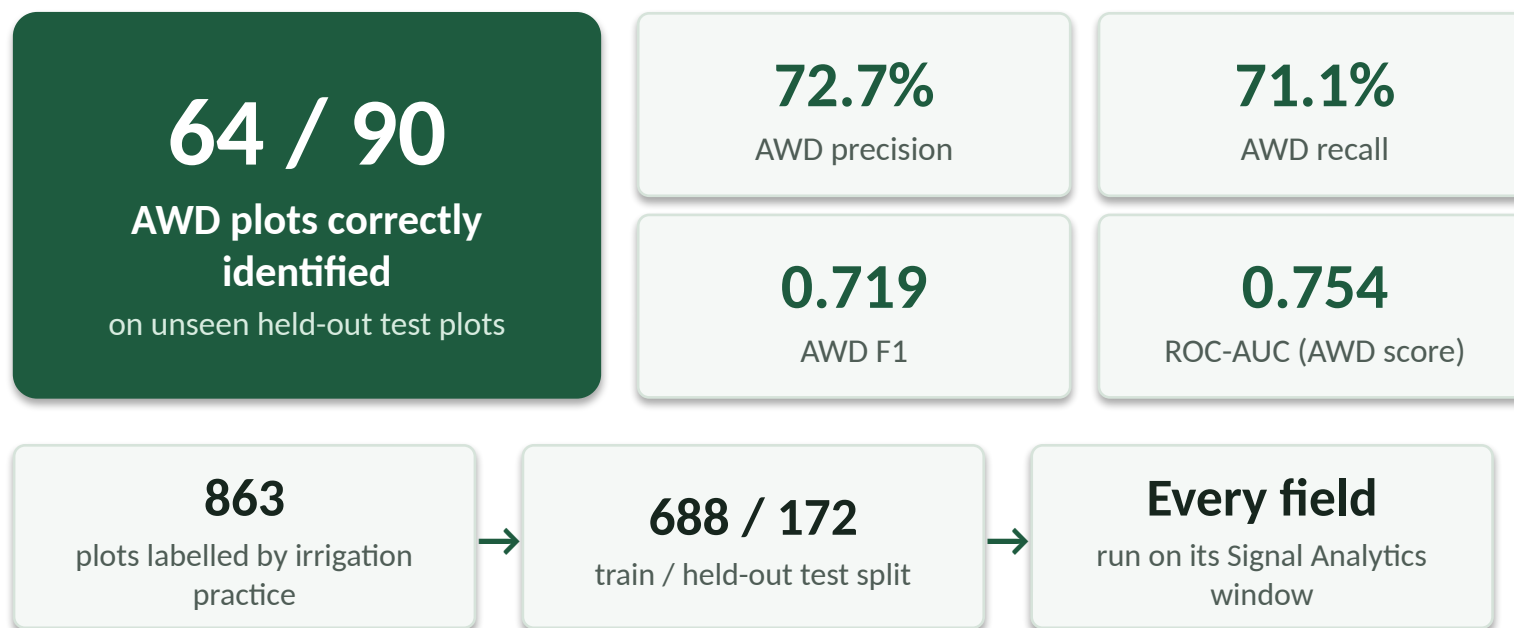
2 AWD events • 84-day season

Live run on the deployed app • Sentinel-1 via Google Earth Engine • 12 observations, Jan–May 2026 • rule engine: AdaptiveAWDGate

A model trained on real AWD plots cross-checks the detector

AWD practice model

Random forest · Sentinel-1 radar + Satellite Embedding · Microsoft rice-irrigation-mapping (Punjab)



Why add ML?

Rules are explainable but brittle. A model trained on independently labelled AWD plots checks each field's drydown result from a second angle.

Honest limitation

Labels come from Punjab, not Bangladesh, and the model is right on about 71% of unseen plots. It stays advisory — it flags disagreement and never changes a calculation.

A deterministic VM0051 engine turns evidence into tCO₂e

$$E = EF_c \times SF_w \times SF_p \times SF_o \times \text{days} \times \text{area}$$

$$\Delta CH_4 = E_{\text{baseline}} - E_{\text{project}}$$

$$\Delta CO_2e = \Delta CH_4 \times 28 \div 1000$$

$$ER = \Delta CO_2e \times (1 - 15\%) - PE_{N_2O}$$

VM0051 Eq. 6–8, 29 (QA3) · $EF_c = 0.85 \text{ kg CH}_4/\text{ha}/\text{day}$ (IPCC 2019, South Asia) · $GWP_{CH_4} = 28$ (AR5)

WATER-REGIME FACTOR SF_w — WHERE AWD'S BENEFIT COMES FROM

1.00

Continuous flooding
0 drydowns

0.71

Single aeration
1 drydown

0.55

True AWD
≥ 2 drydowns

Worked example

1 ha · 84 days · 2 drydowns · 100 kg N/ha

Baseline CH ₄	205.5 kg
Project CH ₄	113.0 kg
Avoided CH ₄	92.5 kg
As CO ₂ e	2.59 t
– 15% uncertainty	– 0.39 t
– N ₂ O penalty	– 0.08 t

Net reduction

2.12 tCO₂e

Above 60,000 tCO₂e/yr the engine blocks instead of guessing.

Worked example computed by CarbonAssetEngine.calculate_credits() (src/carbon/rice.py)

Readiness gates never let an unsupported requirement pass silently

☑️ Readiness checklist — VM0051 field

satisfied

`common.monitoring_period_coverage`
Crop seasons cover the monitoring period

needs review

`common.methodology_applicability`
Declared crop outside usual scope — reviewer confirms

needs review

`common.evidence_review_status`
Field observations not yet accepted by a second person

missing

`vm0051.required_measurement_inputs`
AWD events and season length not yet supplied

unsupported

`vm0051.leakage_assessment`
Not implemented — stays blocking

unsupported

`vm0051.n2o_baseline_fertilizer`
Not implemented — shown to the verifier, never hidden

1 satisfied**3** needs review**1** missing**3** unsupported

Example readiness run on the deployed app (6 of 8 checks shown) · internal approval is not VVB verification or Verra issuance

AT COMMIT, THE SYSTEM FREEZES

- 🔒 Field boundary & area
- 🔒 Crop seasons & engine inputs
- 🔒 Methodology bundle & factors
- 🔒 Readiness result

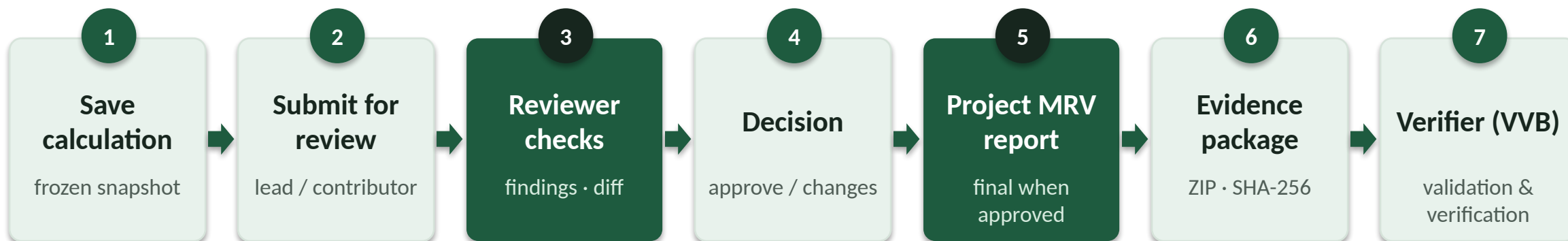
INTERNAL REVIEW

submitted**in_review****internally_approved**

or changes requested / rejected / withdrawn

Internal review turns a calculation into a verifier-ready MRV report

Nothing reaches the report unreviewed — and everything in it can be checked.



Project MRV report (PDF) VM0051 monitoring-report structure

All fields summed, QA3 checked per year, QA/QC trail, requirements, project documents and sign-off.



Evidence package (ZIP) Everything a verifier needs, hashed

Field reports, frozen snapshots, Sentinel-1 results, documents and a SHA-256 manifest of every file.

The screenshot shows the first page of the 'Terra-Audit' Project Monitoring Report for 'New Project'. It includes a table for '1. Fields in this Monitoring Period' and a table for '2. Quantification of Emission Reductions'.

Field	Area (ha)	Monitoring period	GHG emissions	Net GHG emissions	GHG emissions
Forest Farm (forest)	6,400.00	2020-01-01 to 2020-01-01	1	1,000,000	1,000,000
Total	6,400.00			1,000,000	1,000,000

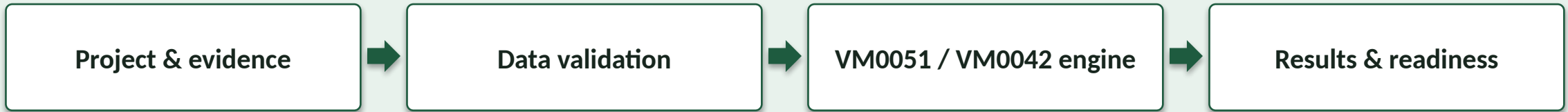
Item	Value (tonnes of CO2e)	Reference
Baseline GHG emissions	1,000,000	Eq. 6.10.1
Project GHG emissions	1,000,000	Eq. 6.10.1
GHG emissions avoided	0	Eq. 6.10.1
Gross emission reductions	0	Eq. 6.10.1
Net emission reductions	0	Eq. 6.10.1

REAL OUTPUT

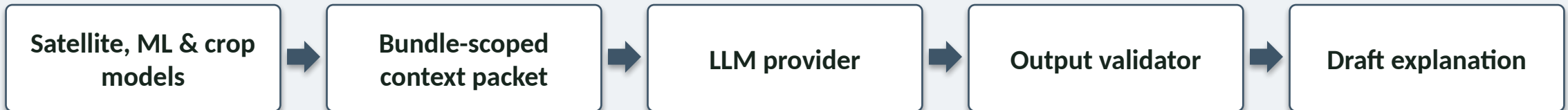
Page 1 of the project report — FINAL once every field is approved.

AI explains, code calculates, humans approve

AUTHORITATIVE • DETERMINISTIC



ADVISORY • AI



 **Bounded context**
AI only sees the active methodology bundle, evidence and calculation.

 **Validated output**
Citations must exist and numbers must match; “certified” or “approved” claims are rejected.

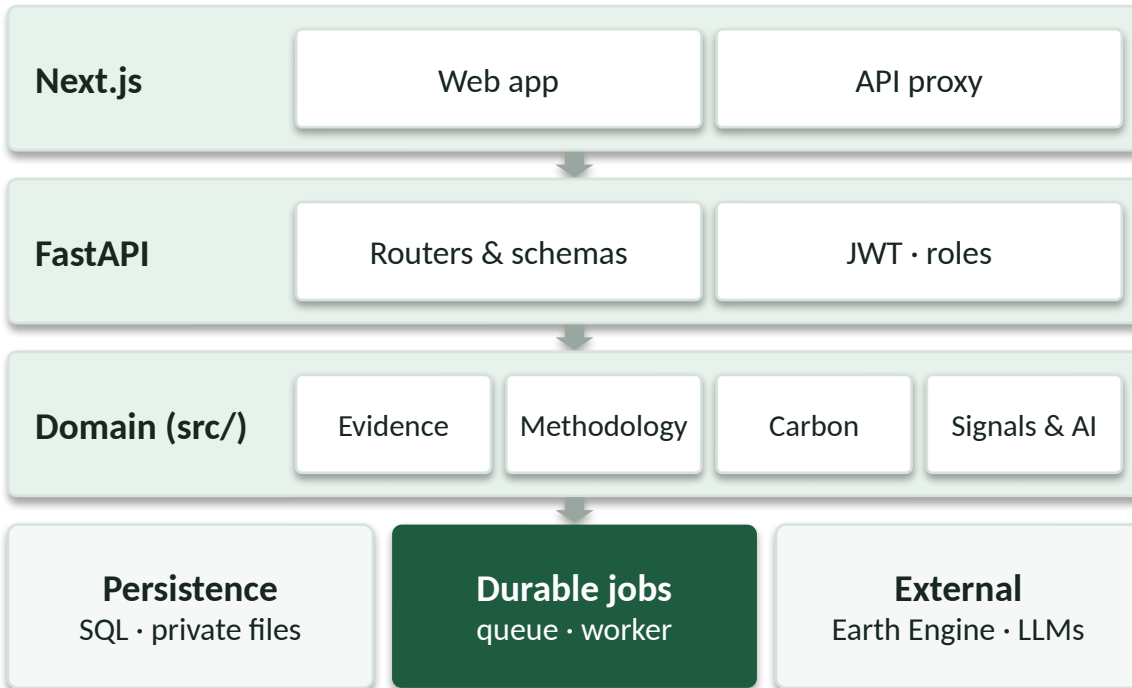
 **Fail closed**
One retry, then fail — nothing partial is saved.


Human reviewer

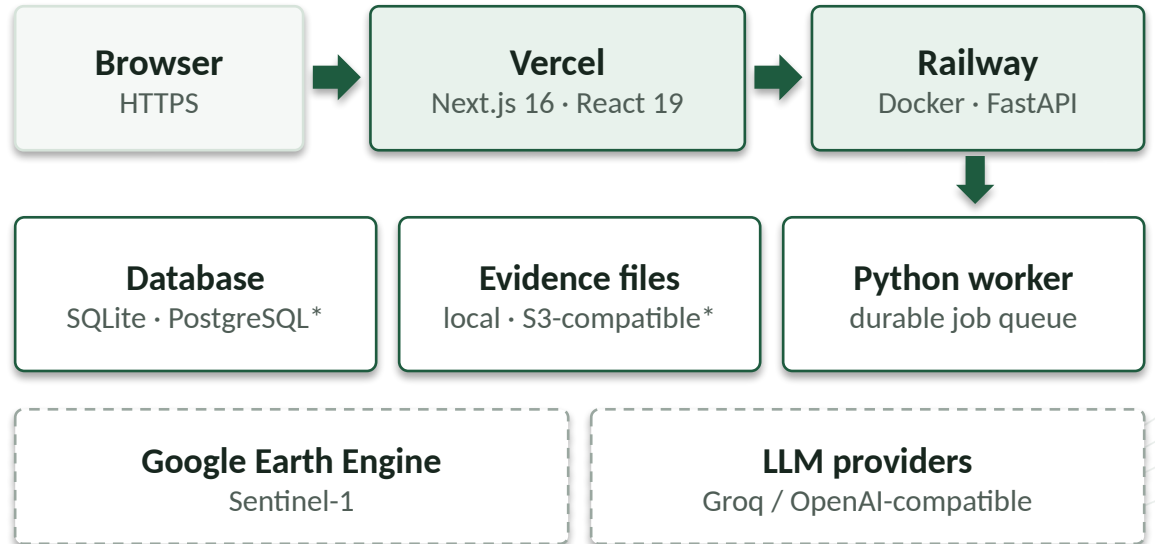
No AI component can change a calculated value, mark a requirement compliant, or create registry credits

Layered components, deployed as web, API and worker

COMPONENT LAYERS



DEPLOYMENT



Solid = verified in the recorded run · dashed = configured external service · * supported hosted option

DESIGN EVOLUTION
midterm → final

Streamlit monolith → **Next.js + FastAPI**

Request-coupled jobs → **Durable queue + worker**

Scalar results → **Versioned snapshots**

Deployment as recorded in the final report (\$5.5) · production worker and storage provenance not yet verified end to end

Design classes, persistent data and behaviour

KEY DESIGN CLASSES & MODULES

AdaptiveAWDGate

analyze_irrigation_behavior()
extract_phenology()

CarbonAssetEngine

calculate_credits()
VM0051 rice AWD

methodology.readiness

build_readiness_checklist()
per-requirement states

WorkerContext · jobs

create_job()
claim_next_job()

SpatialDataEngine

extract_clean_timeseries()
Sentinel-1 via Earth Engine

AlmCarbonEngine

calculate_credits()
VM0042 + VMD0054 leakage

carbon.snapshot

build_snapshot()
calculate_from_snapshot()

AttachmentStorage

LocalFilesystemStorage
S3Storage (strategy)

PERSISTENT DATA (50+ TABLES)



Identity

organizations · users ·
project_members



Fields & evidence

fields · crop_seasons ·
practice_events · attachments



Methodology

methodology_bundles ·
methodology_requirements ·
readiness_determinations



Calculation & review

calculations ·
review_submissions · findings



Jobs & AI

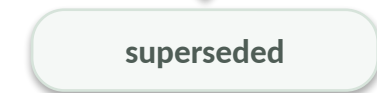
background_jobs · workers ·
ai_records

BEHAVIOUR (STATE MACHINES)

Calculation version

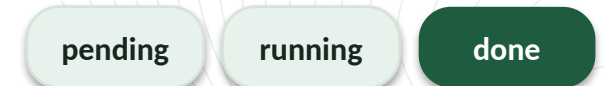


decided by blocking readiness items



new evidence → new version

Background job

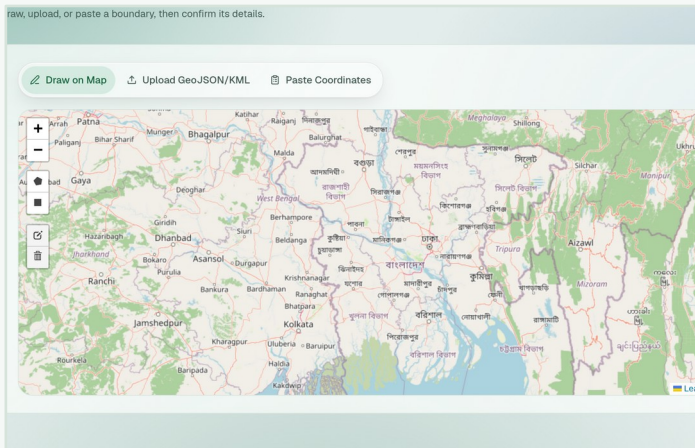


or error (retry / invalid) · cancel_requested →
cancelled · heartbeat-monitored workers

Names as in the source code (src/, backend/worker.py) · review states shown on the readiness slide · full ER model in the report (§4)

Interface design: every user action has a visible state

Register a field



OBJECTS

map canvas • input-mode tabs

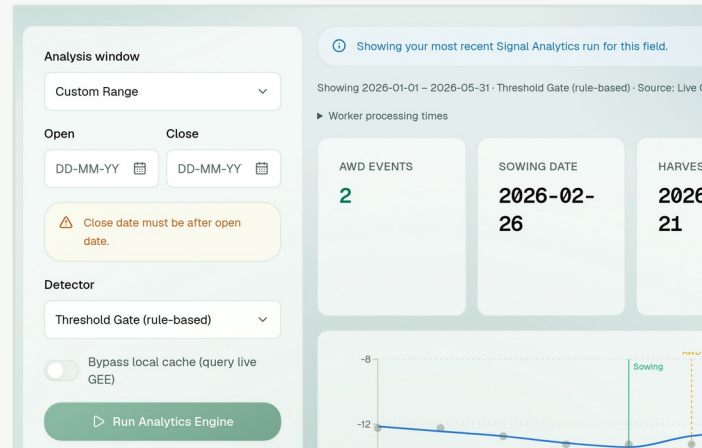
ACTION

draw / upload GeoJSON-KML / paste

STATE

boundary → confirm details → field saved

Signal analytics



OBJECTS

analysis window • detector • Run

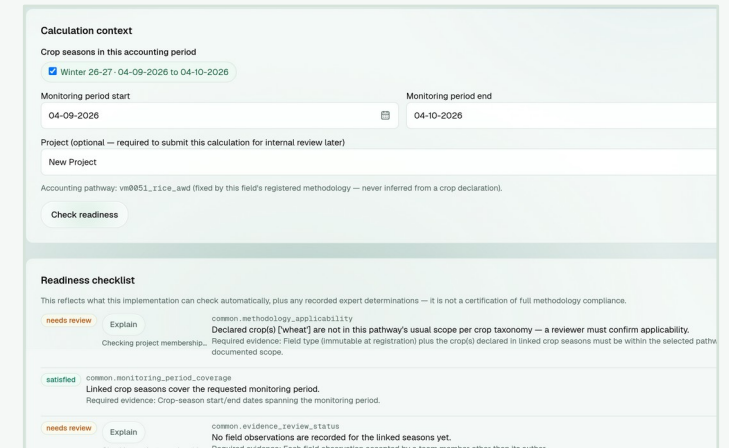
EVENT

close < open → inline error

STATE

job pending → results + audit trail

Calculation & readiness



OBJECTS

seasons • period • Check readiness

EVENT

Check readiness → checklist

STATE

Commit → draft / ready_for_review

Shared labelled controls • inline validation • text + colour status cues • DD-MM-YYYY dates • keyboard-aware controls • responsive layouts

Code: the AWD rule engine produces an auditable trail

Date	VV z-score	Flooded	Drydown
09 Jan	1.023		
21 Jan	0.441		
02 Feb	-0.140		
14 Feb	-0.942	✓	
26 Feb	-1.440	✓	
10 Mar	-0.111		✓
22 Mar	0.390		
03 Apr	-0.606		
15 Apr	-1.326	✓	
27 Apr	-0.083		✓
09 May	0.857		
21 May	1.936		

src/signals/threshold_gate.py · AdaptiveAWDGate (excerpt)

```
# Field-level statistical baseline
vv_mean = df["vv_smoothed"].mean()
vv_std = df["vv_smoothed"].std()
if vv_std == 0 or np.isnan(vv_std):
    vv_std = 0.1 # Prevent division by zero on flat test data

# Z-score flood state detection
df["vv_zscore"] = (df["vv_smoothed"] - vv_mean) / vv_std
df["is_flooded"] = np.where(
    df["vv_zscore"] < self.z_flood_threshold, 1, 0)

# Drydown event = sharp positive jump following a flooded state
df["vv_diff"] = df["vv_smoothed"].diff().fillna(0)
threshold_jump = self.dynamic_delta_sigma * vv_std
df["drydown_event"] = np.where(
    (df["vv_diff"] > threshold_jump)
    & (df["is_flooded"].shift(1) == 1), 1, 0)
```

Left: the columns this code produces, from the live run (audit-trail table in the app) · z_flood_threshold = -0.8, dynamic_delta_sigma = 1.2

Code: the credit engine blocks instead of guessing — and a test proves it

src/carbon/rice.py · calculate_credits() (condensed)

```
# Baseline vs project CH4 (Eq. 6/8)
e_baseline = self.ef_c * 1.0 * sc_p * sc_o_bsl * days * area_ha
e_project  = self.ef_c * sf_w_project * sc_p * sc_o_wp * days * area_ha
delta_e_co2e = (e_baseline - e_project) * self.gwp_ch4 / 1000.0

# QA3 project-size gate (§8.6.3)
if delta_e_co2e > self.QA3_PROJECT_SIZE_GATE_TC02E_YR:
    return {"qa3_pathway_valid": False,
            "final_issuance": None, ...}

# 15% uncertainty, N2O penalty (Eq. 25), net (Eq. 29)
ch4_after_unc = delta_e_co2e * (1 - self.UNC_QA3_DEFAULT)
pe_n2o = (self.n2o_irrigation_penalty(q_n, area_ha)
          if awd_events > 0 else 0.0)
final_issuance = max(0.0, ch4_after_unc - pe_n2o)
```

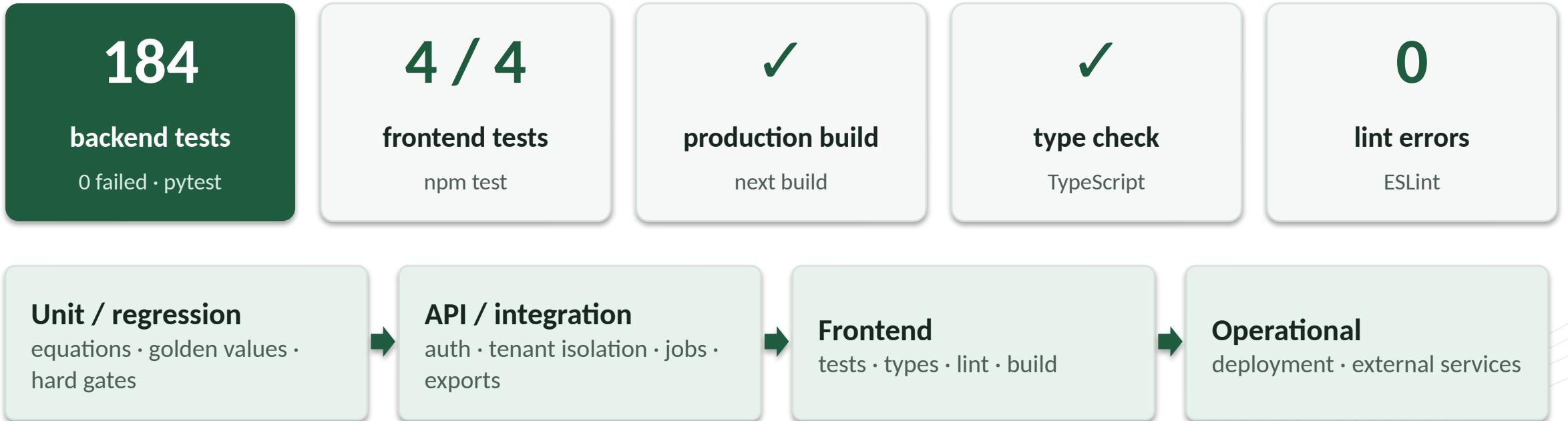
tests/test_carbon_calculator.py

```
def test_qa3_project_size_gate_blocks_oversized_projects(
    engine):
    r = engine.calculate_credits(
        awd_events=2, season_length_days=120,
        area_ha=500_000.0, q_n_kg_per_ha=0.0)
    assert r["qa3_pathway_valid"] is False
    assert r["final_issuance"] is None
```

- Every factor is a named constant cited to its VM0051 / IPCC table
- All intermediate values are returned for the audit trail
- Snapshot calculation never re-reads the live database

Engine code condensed for the slide — full function in src/carbon/rice.py · test passes in the 184-test backend suite

184 backend tests pass — layered from equations to tenant isolation



Pass criterion

0 failures, no unjustified skips — and unsupported states must stay explicit, never silently satisfied.

Risks → contingency

No ground truth → ML advisory · unimplemented rules → blocking checks · live services → isolated fixtures.

All five checks re-run on the current code, 10 Oct 2026 · ESLint: 0 errors, 4 warnings

Key test cases prove gates, isolation and reproducibility

ID	Scenario	Expected	Observed	Result
TC-01	QA3 size gate — 500,000 ha field	Reject above 60,000 tCO ₂ e/yr	qa3_pathway_valid = False, no issuance	PASS
TC-03	Another organization reads a field	Cannot list or open it	empty list · direct GET → 404	PASS
TC-04	PDF after live evidence is edited	Built from the frozen snapshot	owner 200 · viewer 403 · other org 404	PASS
TC-06	AI output claims “certified”	Reject; one retry, then fail	validation error · nothing saved	PASS

Evidence: tests/test_carbon_calculator.py · tests/backend/test_fields_endpoints.py · test_evidence_verification.py · tests/test_ai_validate.py

Challenges solved — and what remains

CHALLENGE → SOLUTION



Methodology fidelity

Factors cited to tables + golden-value tests — caught a wrong EF (1.4 → 0.85).



Slow satellite & AI work

Durable job queue with a separate Python worker.



Numbers drifting after edits

Immutable snapshots; edits create a new version.



STILL OPEN

- VM0051 leakage not yet modelled
- No independent AWD ground truth
- AI evaluation set incomplete
- Production provenance unverified



NEXT

- Field ground-truth campaign
- Eq. 38 uncertainty for large projects
- Remaining VM0051 terms



Thank you

Questions are welcome

Live app terra-audit.vercel.app

Repository github.com/knah1d/terra-audit

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Supervisor Dr. Emon Kumar Dey, Professor, IIT, DU



Live app



GitHub